X-Ray Emitting Plasma in Galaxies and Galaxy Clusters

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May 28, 2026

Abstract

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1 Introduction

In studying the evolving nature of our universe on the largest scales, galaxy clusters represent the most expansive clearly defined structures in which matter approaches virialization. Due to this fact, we can use them as laboratories to study the cosmic growth rate and characterize density fluctuations, as well as reconstruct the dynamical history of merging systems and the effects of feedback from active members. A tenuous plasma called the Intracluster Medium (ICM) permeates these massive objects, with temperatures in the 10^7 K range and densities from 10^{-5} to 10^{-1} cm^{-3} between their outer regions and inner cores. In this hot state, the ICM radiates at several keV photon energies, making spectroscopy in the soft X-ray band a primary method to resolve the distribution of baryonic matter inside clusters (Böhringer & Werner 2010).

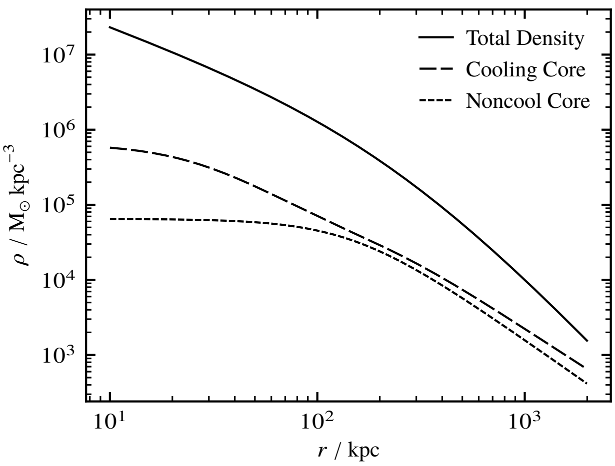

Figure 1: Typical mass density profiles for NCC and CC galaxy clusters are plotted as proxies for all baryonic matter. Shown as well is a standard NFW total density curve that is dominated by a collisionless dark matter component.

1.1 Models

At the broadest level, galaxy clusters are modeled as dark matter halos fully governed by gravitational collapse. They are thought to originate from primordial density fluctuations and grow via cluster mergers as well as by accreting matter from cosmic filaments. Their total mass is dominated by the dark component defined through its purely gravitational interactions, the density of which is well described using a Navarro-Frenk-White (NFW) profile. Most baryonic matter is not contained in the stellar mass of the member galaxies themselves but in the ICM gas, which follows a similar shape to the overall matter distribution in the outer parts where it usually makes up a fraction of roughly $f_g \sim 15\,\%$ depending on redshift, but flattens towards the central region due to counteracting pressures. Containing this material requires a sufficiently deep gravitational potential well that is provided by the high dark matter concentration, which strongly peaks in the center as collisional effects are negligible. Shown in Figure 1 are exemplary radial density profiles with realistic parameter choices for the observed galaxy cluster categories as explained in the following subsections (Pratt et al. 2019).

Adopting this view, radiative cooling of the higher density inner regions through X-ray emission generally leads to a drop in thermal pressure and a resulting influx of subsonic materials from the surrounding medium, called a cooling flow. This assumption predicts massive rates of molecular condensation near the central galaxies which should produce significant starburst activity. However, observations by the *Chandra* (Weisskopf et al. 2000) and *XMM-Newton* (Jansen et al. 2001) missions proved that this signature is not present in reality. Instead, the ICM cools towards the core but does not reach the neutral gas threshold. The leading hypothesis for the culprit of this deviation from theoretical expectations is the injection of energy by Active Galactic Nucleus (AGN) feedback phases regulated by a balancing of heating and fueling, although this description carries with it its own set of open questions, namely the mechanisms for isotropization of highly directional jets as well as stabilization and dissipation of the AGN bubbles (Böhringer & Werner 2010, Rosati et al. 2002, Pratt et al. 2019).

(a) Abell 85

(b) Abell 426 (Perseus)

(c) Abell 1644

(d) Abell 1656 (Coma)

(e) Abell 2029

(f) Abell 3158

Figure 2: Selection of galaxy clusters used as examples in this work. Displayed here are composite overlays of second generation *Digitized Sky Survey* red band images and contours for *ROSAT All Sky Survey (RASS)* broad band count rates (Truemper 1982, Lasker 1994). Both optical and X-ray data are stretched using a simple 0.8 exponent power law. Each frame has a radius of $15'$ for its field of view. Refer to Figure 9 for raw *ROSAT* counts underlying the contours.

1.2 Classes

According to the spherical dark matter halo assumption, gas properties should follow selfsimilar laws along the radial distance to the cluster center and only scale with the total mass and redshift. Comparing to observations such as those presented in Figures 7 and 8 shows that this holds quite well for large radii but breaks down closer to the central regions. It turns out that based on these differences, galaxy clusters can be sorted into two categories. In order to do so, it is useful to introduce the cooling time. It derives from the total thermal energy $E_t \propto nT$ and volumetric cooling rate $\dot{E}_c \propto n^2 \Lambda_c(T)$ where the cooling function for bremsstrahlung scales as $\Lambda_c(T) \propto T^{1/2}$ to define a proportionality

$$t_c = \frac{E_t}{\dot{E}_c} \propto T^{1/2} n^{-1}$$

for the ICM properties. If this timescale is small $t_c < 1\,\mathrm{Gyr}$ in its center, we call it a Cooling Core (CC) cluster. These systems radiate away heat efficiently, leading to a rapid drop in gas temperature and a corresponding peak in density. With these characteristics, they also have an almost critically low entropy floor, close to the catastrophic cooling flow collapse scenario and only barely stabilized by galactic winds from star formation and AGN feedback. Generally, the population of CC clusters comprises dynamically relaxed systems with a high degree of order. In the opposite case, cooling is highly inefficient with $t_c > 10\,\mathrm{Gyr}$ and frequently exceeding the Hubble time. Such clusters are labeled Noncool Core (NCC) and tend to possess isothermal or monotonically increasing temperatures and flat density profiles towards their centers. Their inner entropy floors are large and likely originate from merging of subclusters. Heating occurs through supersonic shocks, turbulence, contact discontinuities and sloshing of the central material.

From this follows that analysis of the thermal structure in principle allows us to infer the recent dynamical history of clusters. Additionally, measuring the total mass M inside some characteristic radius r makes it possible to map the distribution of large scale density inhomogeneities and their evolution throughout cosmological time. After determining plasma properties and by assuming hydrostatic equilibrium, we can rearrange the well known formula

$$dP/dr = g\rho = GM\rho/r^2$$

to solve for mass. A common choice for a universal distance normalization is the overdensity radius R_ℓ which constrains the sphere around the cluster center inside which the average mass density is a factor ℓ larger than the surrounding critical density of the universe at that redshift. This way, contained masses are labeled M_ℓ and average quantities T_ℓ or P_ℓ as well as astrophysical entropy K_ℓ with $K \propto Tn^{-2/3}$ can be defined for comparison between systems of vastly different sizes. Typical values are $\ell = 200$ since R_{200} is close to the theoretical virialization radius, as well as $\ell = 500$ and R_{500} as the boundary up to which current X-ray telescopes can produce reliable measurements (Böhringer & Werner 2010, Pratt et al. 2019).

1.3 Examples

To have a consistent set of objects to compare theoretical predictions to real observations for this work, we select six clusters. Their optical appearance as well as X-ray surface brightness contours are depicted in Figure 2 with Table 1 listing their characterizing properties.

Table 1: Parameters for our galaxy cluster sample. Values given are those determined for the *X-COP* systems (Eckert et al. 2017).

Name	Class	z	R_{500} / kpc	M_{500} / $M_\odot$
A85	CC	0.0555	1200	5.6×10^{14}
A426	CC	0.0179	1300	5.9×10^{14}
A1644	NCC	0.0473	2100	3.5×10^{14}
A1656	NCC	0.0231	1300	6.0×10^{14}
A2029	CC	0.0766	1400	8.6×10^{14}
A3158	NCC	0.0590	1100	4.3×10^{14}

The Perseus or Abell 426 as well as the massive Abell 2029 clusters are archetypical examples of relaxed CC systems supported by regulated feedback loops, although there are still remnants of ancient disruptions detectable. Abell 85 also falls in the CC category but shows more prominent central sloshing as well as active accretion from intersecting cosmic filaments. The Coma or Abell 1656 cluster on the other hand is a prototype for NCC systems that have been disrupted by a recent massive merger event. Similarly, high resolution observations classify Abell 3158 as NCC despite its smooth X-ray surface. Lastly, Abell 1644 is clearly an NCC cluster as is readily apparent from its bimodal halo structure (Böhringer & Werner 2010, Eckert et al. 2017).

2 Radiation Mechanisms

While we can assume Local Thermodynamic Equilibrium (LTE) for thick plasmas such as those inside stars, where excitation obeys the Boltzmann distribution and ionization the Saha equation with radiation following a Planck law, the ICM is thin except for some resonant scattering effects. This means that photons escape easily, leaving the balance of ionization and recombination to be regulated entirely through the plasma particles themselves, in which electrons retain their Maxwellian velocity distribution. We refer to such systems as being in Collisional Ionization Equilibrium (CIE) which have certain characteristic radiation properties (Kaastra et al. 2008, Rybicki & Lightman 1979).

2.1 Free-Free Emission

When free electrons get deflected by the Coulomb field of an ion, they lose kinetic energy via the emission of photons. Since the amplitude and occurrence rate are dependent on the state of the thermalized plasma population, we call this mechanism thermal braking radiation or bremsstrahlung. Its emissivity is given by the equation

$$\epsilon \propto n_e n_i Z^2 T^{-1/2} g_{ff}(T, E) e^{-E/kT}$$

where the Gaunt factor $g_{ff}(T, E)$ is an averaged quantum mechanical correction of order unity. The spectra produced by bremsstrahlung present a characteristic broad continuum with an exponential rolloff once photon energies exceed the order of the thermal electron energy. This cutoff from the Gaunt factor dominated flat slope can therefore be used to determine the local temperature of an emitting region, as is shown in Figure 3 for different cases.

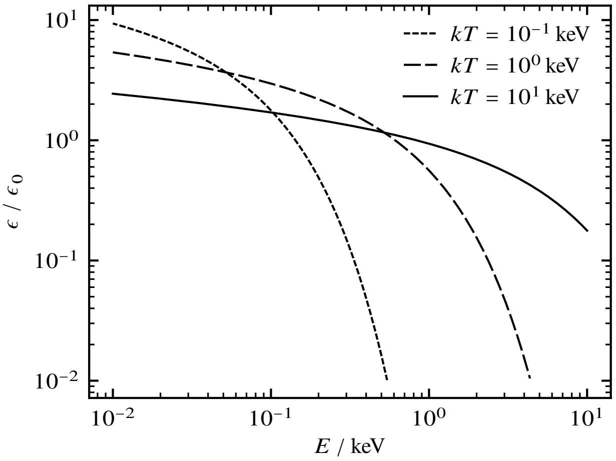

Figure 3: Normalized bremsstrahlung continua for different temperatures over a range of magnitudes. Fitting the bend placement allows for the inference of local temperature.

Since the ICM is very hot with thermal energies of the order 10 keV it is almost entirely dominated by the bremsstrahlung spectrum. However, systems with shallower gravitational potentials and lower temperatures than clusters are often still in CIE and exhibit a less prominent continuum, leading to other radiation components being more relevant. Examples for this are galaxy groups as well as giant elliptical galaxies (Böhringer & Werner 2010).

2.2 Bound-Free Component

If a free electron gets captured into a bound state by an ion, it emits photons with a frequency corresponding to the sum of its previous kinetic energy and the ionization potential of the specific atomic shell. This results in a spectrum defined by a sharp recombination edge at the minimum required energy for ionization of the excitation level and a smooth tail toward higher energies. Since lighter elements like hydrogen and helium are fully stripped of electrons, their contributions to the ICM radiation are negligible, although heavier metals like iron can show up as subtle steps in the bremsstrahlung continuum. At lower temperatures, this mechanism becomes more visible as ions retain more bound electrons.

2.3 Bound-Bound Features

Emission lines result from collisional or radiative excitation and the subsequent decay of an electron between two bound energy states. Due to the quantization of atomic shells, the emitted photons have a very narrow energy range. Realistic line shapes are not perfectly discrete because of the intrinsic Heisenberg uncertainty and can be further broadened by the thermal Doppler effect or deformation from collisions. From convolving Lorentzian and Gaussian curves, they obtain a Voigt profile depending on the plasma motion. Since in the ICM most atomic nuclei are stripped bare with no bound electrons left, few if any spectral lines are present in their bremsstrahlung continua. For the analysis of galaxy clusters through X-ray spectroscopy, the most important ones are the Fe K complex centered at 6.6 keV and the Fe L complex around 1.1 keV as well as O lines near 0.6 keV for reasons discussed in the following sections (Rybicki & Lightman 1979, Kaastra et al. 2008, Böhringer & Werner 2010).

2.4 Sunyaev-Zeldovich Effect

Although it is observed in the microwave and submillimeter bands of the electromagnetic spectrum instead of at X-ray energies, the thermal Sunyaev-Zeldovich (SZ) effect offers a highly relevant complimentary approach for mapping ICM properties. It describes how Cosmic Microwave Background (CMB) photons are boosted towards higher energies through inverse Compton scattering with energetic ICM electrons. This produces a shadow below and an enhancement above a specific frequency versus the nearly homogenous CMB blackbody spectrum. Since the energy density at larger z is higher by a factor exactly equal to the dimming magnitude during travel of the signal, the SZ effect does not depend on redshift. As it involves one body, its surface brightness scales linearly with $P \propto nT$ as opposed to the square dependence in $S \propto n^2 T^{1/2}$ for bremsstrahlung from the interaction of two charges (Birkinshaw 1999, Böhringer & Werner 2010, Rybicki & Lightman 1979). Combining measurements of the ICM emission itself in X-rays and SZ cluster maps from *Planck* data allows us to build high quality models of their gas structure (Aghanim et al. 2020). Visualized in Figure 4 is a comparison to the CMB spectrum.

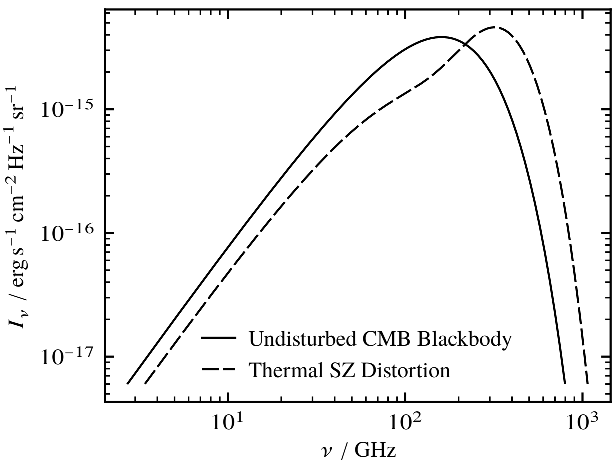

Figure 4: Schematic view of the thermal SZ effect on the CMB with exaggerated $Y \sim 0.2$ Compton parameter.

3 Instrumentation

By conducting the first sensitive all sky X-ray survey and building catalogs of galaxy clusters, the *Röntgen Satellite (ROSAT)* mission paved the way for following instruments (Truemper 1982). Currently, the two most established active telescopes are *Chandra* and *XMM-Newton* (Weisskopf et al. 2000, Jansen et al. 2001). While the National Aeronautics and Space Administration (NASA) operated *Chandra* can achieve very high spatial resolutions with its Advanced CCD Imaging Spectrometer (ACIS) and High Resolution Camera (HRC) at below 1″ angular precision, the European Space Agency (ESA) run *XMM-Newton* observatory prioritizes sensitivity using its Reflection Grating Spectrometer (RSG) and European Photon Imaging Camera (EPIC) instruments. Recently launched ongoing missions include the *Spectrum Röntgen Gamma (SRG)* satellite which carries the eROSITA instrument for compiling catalogs and conducting wide field surveys of cosmic filaments in the soft X-ray band, as well as the *X-Ray Imaging and Spectroscopy Mission (XRISM)* headed by the Japan Aerospace Exploration Agency (JAXA) that uses a cooled microcalorimeter array and wide field CCD imager to reach unmatched spectral resolutions for the study of particle and enrichment dynamics in cluster cores (Predehl et al. 2021, Tashiro et al. 2024, Zhang et al. 2026). Perhaps the most capable upcoming mission is *NewAthena* which will feature a massive cryogenic microcalorimeter array with its X-Ray Integral Field Unit (XIFU) to achieve extremely resolved spectra over large patches of clusters and filaments, accompanied by the Wide Field Imager (WFI) camera for deep surveys (Cruise et al. 2025).

4 Spectral Characteristics

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4.1 Continuum

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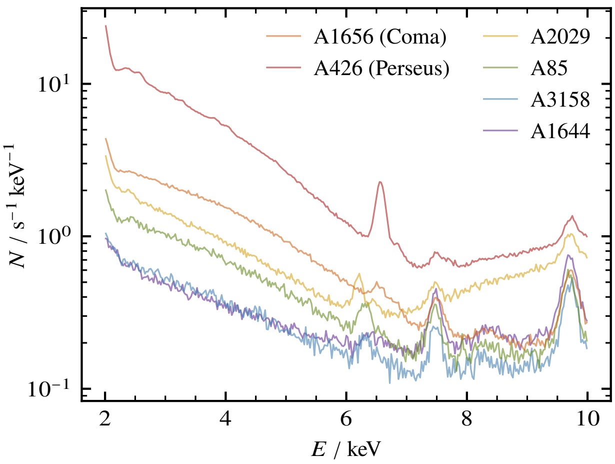

Figure 5: Binned raw *Chandra* spectra for cluster selection with uncorrected systematics (Weisskopf et al. 2000).

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4.2 Line Emission

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5 Plasma Diagnostics

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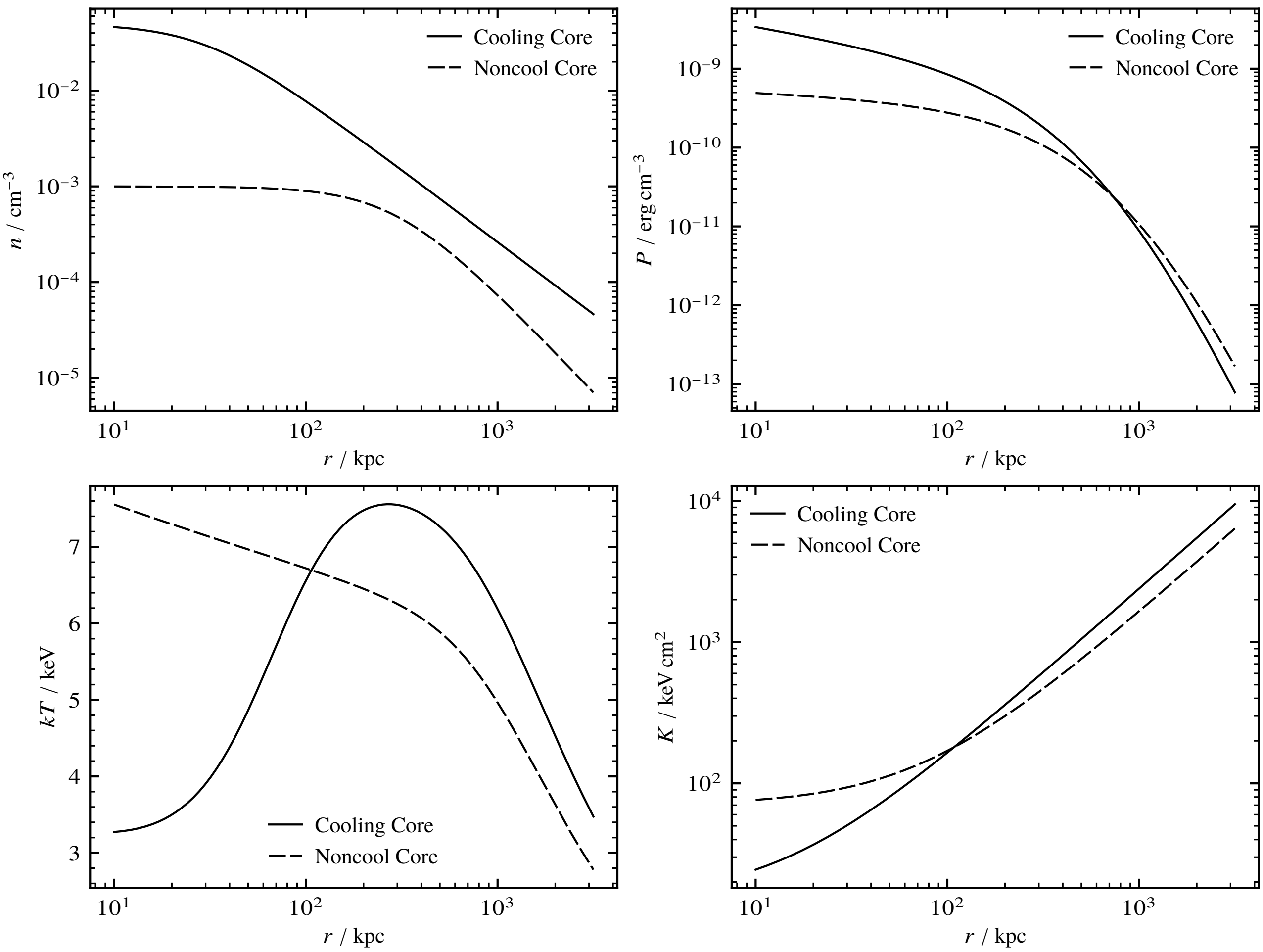

Figure 6: Theoretical galaxy cluster radial profiles assuming universal selfsimilar scaling relations. *Top Left:* Electron number density (Vikhlinin et al. 2006). *Top Right:* Gas pressure (Arnaud et al. 2010). *Bottom Left:* Gas temperature (Vikhlinin et al. 2006). *Bottom Right:* Gas entropy (Cavagnolo et al. 2009). Compare to Figure 7 for real observations.

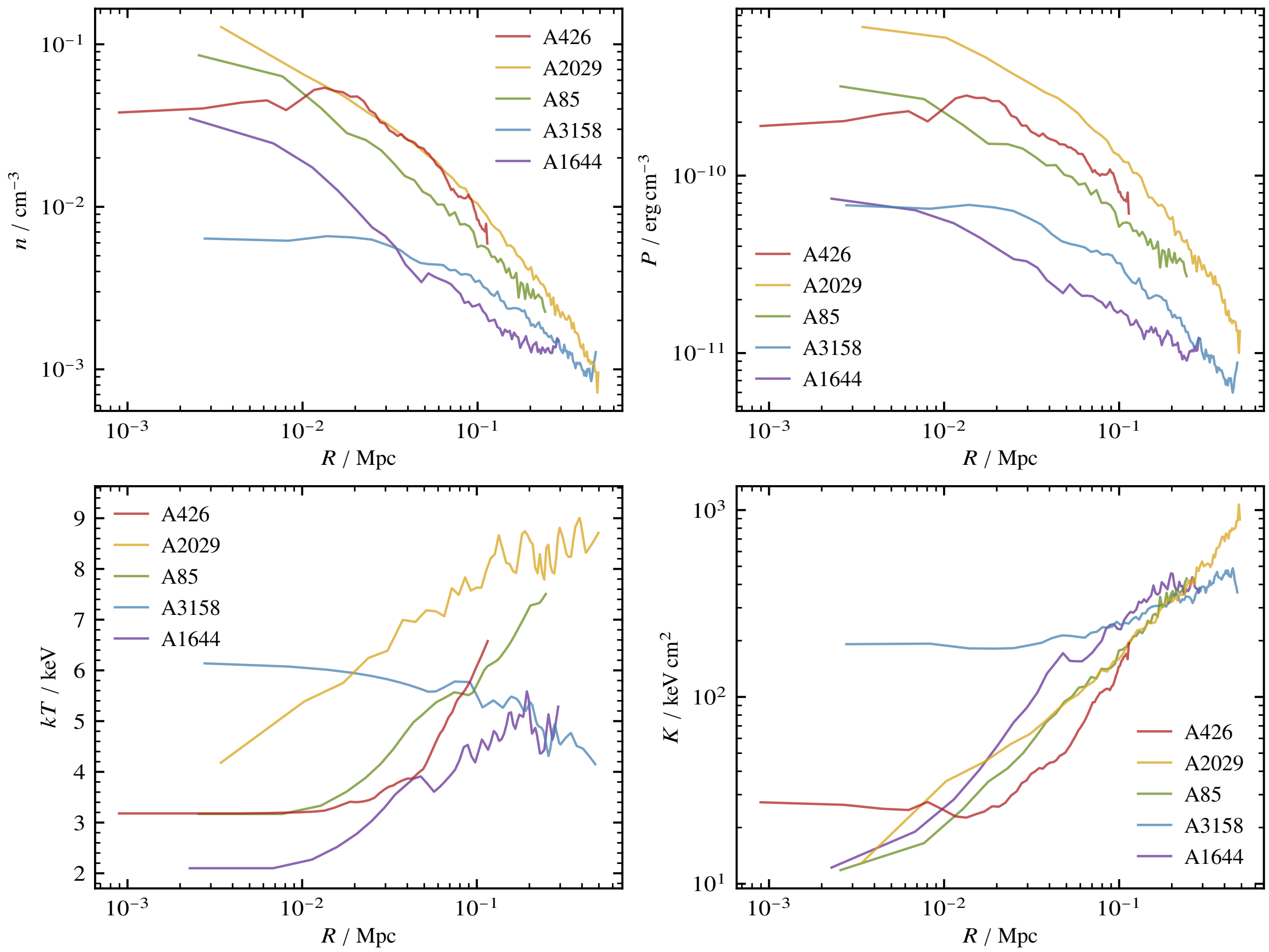

Figure 7: Sample of galaxy cluster radial profiles from *Chandra* observations (Weisskopf et al. 2000). *Top Left:* Electron number density. *Top Right:* Gas pressure. *Bottom Left:* Gas temperature. *Bottom Right:* Gas entropy. Compare to Figure 6 for a visualization of theoretical and empirical fits with realistic parameter choices.

6 Conclusion

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References

Aghanim, N. et al. *Astronomy and Astrophysics* (2020)
Arnaud, M. et al. *Astronomy and Astrophysics* (2010)
Birkinshaw, M. *Physics Reports* (1999)
Böhringer, H. & Werner, N. *The Astronomy and Astrophysics Review* (2010)
Cavagnolo, K. W. et al. *The Astrophysical Journal Supplement* (2009)
Cruise, M. et al. *Nature Astronomy* (2025)
Eckert, D. et al. *Astronomische Nachrichten* (2017)
Jansen, F. et al. *Astronomy and Astrophysics* (2001)
Kaastra, J. S. et al. *Space Science Reviews* (2008)
Lasker, B. M. *Bulletin of the American Astronomical Society* (1994)
Pratt, G. W. et al. *Space Science Reviews* (2019)
Predehl, P. et al. *Astronomy and Astrophysics* (2021)
Rosati, P. et al. *Annual Review of Astronomy and Astrophysics* (2002)
Rybicki, G. B. & Lightman, A. P. *Radiative Processes in Astrophysics* (1979)
Tashiro, M. et al. *Proceedings of the Society of Photo Optical Instrumentation Engineers* (2024)
Truemper, J. *Advances in Space Research* (1982)
Vikhlinin, A. et al. *The Astrophysical Journal* (2006)
Weisskopf, M. C. et al. *Proceedings of the Society of Photo Optical Instrumentation Engineers* (2000)
Zhang, C. et al. *Astronomy and Astrophysics* (2026)

Appendix

A Methodology

B Cluster Profiles

C Reference Images

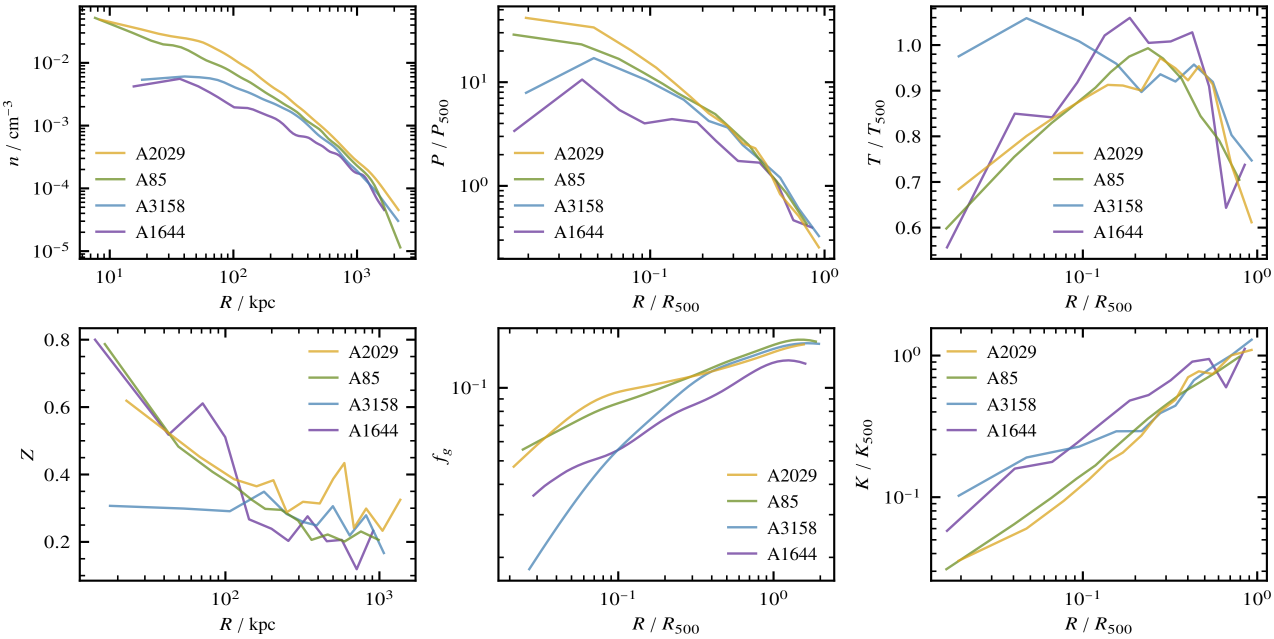

Figure 8: Plots of extracted radial profile projections for available *X-COP* galaxy clusters. *Upper Left:* Electron number density. *Upper Middle:* Gas pressure. *Upper Right:* Gas temperature. *Lower Right:* Gas entropy. *Lower Middle:* Gas mass fraction. *Lower Left:* Metal abundance. Consult the text for definitions of characteristic quantities and scaling. Compare as well to reconstructed distributions in Figure 7 which focus on the inner regions instead of the cluster outskirts.

(a) Abell 85

(b) Abell 426 (Perseus)

(c) Abell 1644

(d) Abell 1656 (Coma)

(e) Abell 2029

(f) Abell 3158

Figure 9: Depicted are *RASS* counts used for X-ray contours shown in Figure 2 after applying a stretch and Gaussian blur (Truemper 1982). The sampled clusters are selected based on brightness and morphology, as well as data accessibility in the *Archive of Chandra Cluster Entropy Profile Tables (ACCEPT)* and *XMM Cluster Outskirts Project (X-COP)* catalogs (Cavagnolo et al. 2009, Eckert et al. 2017).